

Definitions of probability

Classical, statistical and geometric — three definitions, one number between 0 and 1.

LESSON 94–96

3 × 40 MIN

ALGEBRA 10, §30

IGX 24.2–24.4

LESSON CLOCK

25'

25'

25'

35'

20'

5

The classical definition	25 min
Relative frequency	25 min
Geometric probability	25 min
The two rules	35 min
Conditional probability	20 min
Homework	5 min

LEARNING OBJECTIVES

- Use the classical definition $P(A) = m/n$ for equally likely outcomes.
- Estimate a probability from relative frequency, and say when that is necessary.
- Use the addition and multiplication rules, and the complement.
- Compute a simple conditional probability from a table or a tree.

TEXTBOOK

National	pp. 319–334
Cambridge	Core & Extended, pp. 629–642
Workbook	Exercise 24.2–24.4

01

Explanation

The classical definition

When every outcome of a trial is equally likely, probability is a count divided by a count:

$$P(A) = \frac{m}{n} = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

Every probability lies between 0 and 1, with $P(\varnothing) = 0$ and $P(U) = 1$.

EVENT	M	N	P
a die shows 3	1	6	$\frac{1}{6}$
a die shows an even number	3	6	$\frac{1}{2}$
two dice total 7	6	36	$\frac{1}{6}$
a card is a heart	13	52	$\frac{1}{4}$
two coins give two heads	1	4	$\frac{1}{4}$

“EQUALLY LIKELY” IS A CONDITION, NOT A DECORATION

A drawing pin lands point-up or point-down, but not with probability $\frac{1}{2}$. The classical definition applies only when symmetry guarantees equal outcomes; otherwise you must measure.

Relative frequency

When outcomes are not equally likely — a bent coin, a bus arriving late, a component failing — probability is **estimated** from experiment:

$$P(A) \approx \frac{\text{number of times A happened}}{\text{number of trials}}$$

This is the **statistical** definition. The estimate improves as the number of trials grows, and settles down towards a fixed number — the law of large numbers, seen but not proved here.

THROWS	HEADS	RELATIVE FREQUENCY
10	7	0.700
100	54	0.540
1000	509	0.509
10 000	5023	0.5023

Expected number. If $P(A) = p$ and the trial is repeated N times, the expected number of successes is Np . Rolling a die 300 times, expect $300 \times \frac{1}{6} = 50$ sixes — expect, not guarantee.

Geometric probability

When the outcomes form a continuous region, counting is replaced by measuring:

$$P(A) = \frac{\text{measure of the favourable region}}{\text{measure of the whole region}}$$

The measure is a length, an area or a volume, whichever the problem lives in.

PROBLEM	MEASURE	P
a point on $[0, 10]$ falls in $[3, 5]$	length	0.2
a dart in a square of side 4 hits the inscribed circle	area	$\frac{\pi}{4} \approx 0.785$
a bus is due any time in a 20-minute window; you wait 5	length	0.25

A SINGLE POINT HAS PROBABILITY ZERO

Landing at exactly $x = 3$ has probability 0 — yet it is not impossible. On a continuous space, probability zero and impossible are different things.

The two rules

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Subtract the overlap, or it is counted twice. When A and B are mutually exclusive the overlap is empty and the rule shortens to $P(A) + P(B)$.

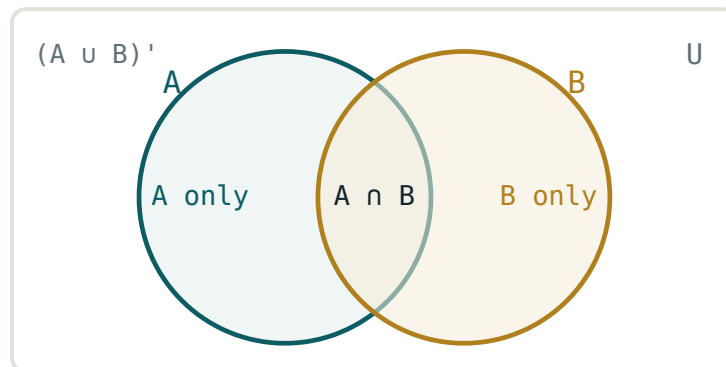
$$P(A \cap B) = P(A) \cdot P(B) \text{ for independent events}$$

$$P(A') = 1 - P(A)$$

THE COMPLEMENT IS THE SHORTEST ROUTE

“At least one six in four throws” has many favourable cases; “no six in four throws” has one description.

So $P = 1 - \left(\frac{5}{6}\right)^4 \approx 0.518$. Whenever a question says **at least**, try the complement first.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The overlap subtracted once — the whole of the addition rule in one picture.

Conditional probability

$P(B | A)$ is the probability of B **given that** A has happened. Conditioning shrinks the sample space to A :

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

From a table. Of 30 students, 18 study French, 12 of whom also study German.

Then $P(\text{German} | \text{French}) = \frac{12}{18} = \frac{2}{3}$ — the denominator is 18, not 30. That single change of denominator is the whole idea.

From a tree. The numbers on the second row of forks are already conditional probabilities. A bag of 3 red and 2 blue, drawn without replacement: $P(\text{second red} | \text{first red}) = \frac{2}{4}$.

$P(B | A)$ AND $P(A | B)$ ARE DIFFERENT NUMBERS

Most people who are ill have a temperature; most people with a temperature are not seriously ill. Swapping the two is the commonest error in the whole of probability.

02 Worked examples

EXAMPLE 1 Two dice are thrown. Find the probability that the total is 7 or 11.

- ① Total 7: 6 outcomes; total 11: 2.

Exclusive events.

②
$$P = \frac{6}{36} + \frac{2}{36}$$

③
$$= \frac{8}{36} = \frac{2}{9} \approx 0.222$$

ANSWER

$$\frac{2}{9}$$

EXAMPLE 2 A coin is thrown four times. Find the probability of at least one head.

- ① Complement: no head at all.

②
$$P(\text{no head}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

③
$$P(\text{at least one}) = 1 - \frac{1}{16} = \frac{15}{16}$$

ANSWER

$$\frac{15}{16} = 0.9375$$

EXAMPLE 3 A bag holds 3 red and 2 blue balls. Two are drawn without replacement. Find $P(\text{both red})$ and $P(\text{one of each})$.

$$\textcircled{1} \quad P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

Multiply along the branch.

$$\textcircled{2} \quad P(RB) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

$$\textcircled{3} \quad P(BR) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$$

$$\textcircled{4} \quad P(\text{one of each}) = \frac{3}{5}$$

Add down the ends.

ANSWER

$$\frac{3}{10} \text{ and } \frac{3}{5}$$

EXAMPLE 4 Of 40 pupils, 25 play football, 18 play chess and 10 play both. Find $P(\text{football} \mid \text{chess})$.

$$\textcircled{1} \quad P(F \cap C) = \frac{10}{40}$$

$$\textcircled{2} \quad P(C) = \frac{18}{40}$$

$$\textcircled{3} \quad P(F \mid C) = \frac{10}{18} = \frac{5}{9}$$

The denominator shrinks to 18.

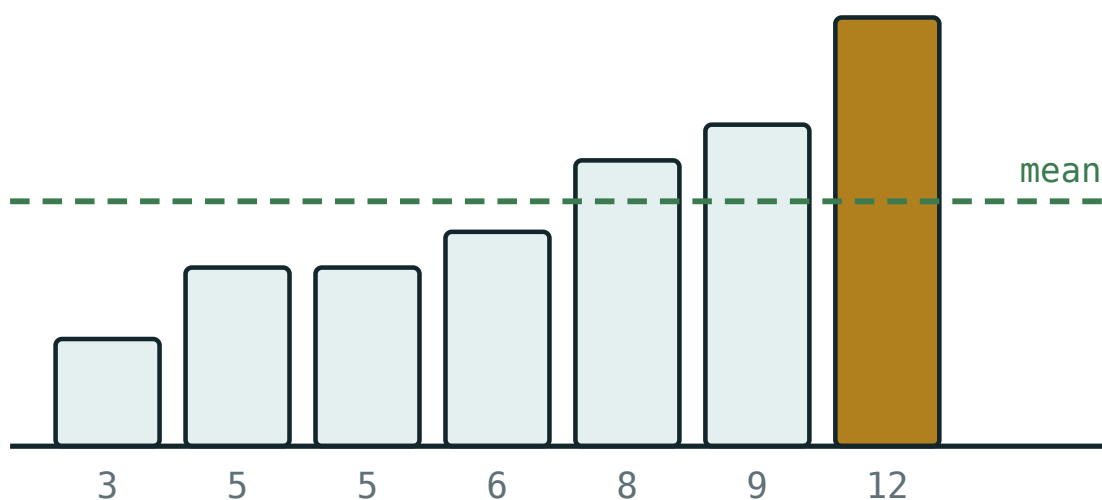
ANSWER

$$\frac{5}{9} \approx 0.556$$

03 Interactive model

Throw one coin 100 times as a class and plot the running relative frequency — the curve settling on 0.5 is the lesson.

Frequency, and what it settles on



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Classical definition	Klassik ta'rif	Классическое определение
Relative frequency	Nisbiy chastota	Относительная частота
Statistical probability	Statistik ehtimollik	Статистическая вероятность
Geometric probability	Geometrik ehtimollik	Геометрическая вероятность
Addition rule	Qo'shish qoidasi	Правило сложения
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Conditional probability	Shartli ehtimollik	Условная вероятность
Complement	To'ldiruvchi	Дополнение
Expected number	Kutilayotgan son	Ожидаемое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The classical definition needs the outcomes to be:

few

equally likely

independent

exclusive

$P(A')$ equals:

$1 - P(A)$

$P(A)$

$1 + P(A)$

0

For exclusive events, $P(A \cup B)$ is:

$P(A)P(B)$

$P(A) + P(B)$

$P(A) - P(B)$

1

For independent events, $P(A \cap B)$ is:

$$P(A) + P(B)$$

$$P(A)P(B)$$

$$0$$

$$1$$

“At least one” is usually done by:

listing every case

the complement

a tree with ten rows

geometry

$P(B | A)$ has denominator:

$$P(B)$$

$$P(A)$$

$$1$$

$$P(A \cup B)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. P a die shows 5

ANSWER $\frac{1}{6}$

2. P a die shows an odd number

ANSWER $\frac{1}{2}$

3. P a card is a heart

ANSWER $\frac{1}{4}$

4. P two coins give two tails

ANSWER $\frac{1}{4}$

5. $P(A')$ when $P(A) = 0.3$

ANSWER 0.7

6. Expected sixes in 300 throws of a die

ANSWER 50

7. P a point of $[0, 10]$ lies in $[3, 5]$

ANSWER 0.2

Medium · the standard question

7 PROBLEMS

1. P two dice total 7

ANSWER $\frac{1}{6}$

2. P two dice total 7 or 11

ANSWER $\frac{2}{9}$

3. P at least one head in three throws

ANSWER $\frac{7}{8}$

4. $P(A \cup B)$ when $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$

ANSWER 0.7

5. Bag of 3 red, 2 blue, two drawn without replacement: $P(RR)$

ANSWER $\frac{3}{10}$

6. Same bag: P one of each colour

ANSWER $\frac{3}{5}$

7. A dart in a square of side 4 hits the inscribed circle

ANSWER $\frac{\pi}{4} \approx 0.785$

Hard · stretch and reason

7 PROBLEMS

1. P at least one six in four throws of a die

ANSWER $1 - \left(\frac{5}{6}\right)^4 \approx 0.518$

2. Of 40 pupils, 25 play football, 18 chess, 10 both: $P(F | C)$

ANSWER $\frac{5}{9}$

3. Same data: P a pupil plays neither

ANSWER $\frac{7}{40}$

4. Two cards from 52 without replacement: P both aces

ANSWER $\frac{1}{221}$

5. Three coins: P exactly two heads

ANSWER $\frac{3}{8}$

6. A bus is equally likely at any time in 20 minutes; you arrive and wait at most 5: P

ANSWER 0.25

7. If $P(A) = 0.6$, $P(B) = 0.5$ and they are independent, find $P(A \cup B)$

ANSWER 0.8

07 Homework

Homework — 5 tasks

State which definition you are using — classical, statistical or geometric — in every answer.

- Two dice are thrown. Find the probability that the total is 9, and that the total is at least 10.
- A coin is thrown five times. Find the probability of at least one tail.

3. A bag holds 4 white and 6 black balls; two are drawn without replacement. Find the probability that they are the same colour.
4. In a class of 30, 20 study English, 14 study Russian and 8 study both. Find $P(\text{English} \mid \text{Russian})$ and $P(\text{Russian} \mid \text{English})$, and comment on the difference.
5. A point is chosen at random inside a square of side 6. Find the probability that it is within 1 unit of an edge.